(a) Using the substitution $u = \sqrt{2x+1}$, show that

$$\int_4^{12} \sqrt{8x+4}\, e^{\sqrt{2x+1}}\, dx$$

may be expressed in the form

$$\int_a^b ku^2 e^u\, du$$

where a, b and k are constants to be found.

(4)

(b) Hence find, by algebraic integration, the exact value of

$$\int_4^{12} \sqrt{8x+4}\, e^{\sqrt{2x+1}}\, dx$$

giving your answer in simplest form.

(5)